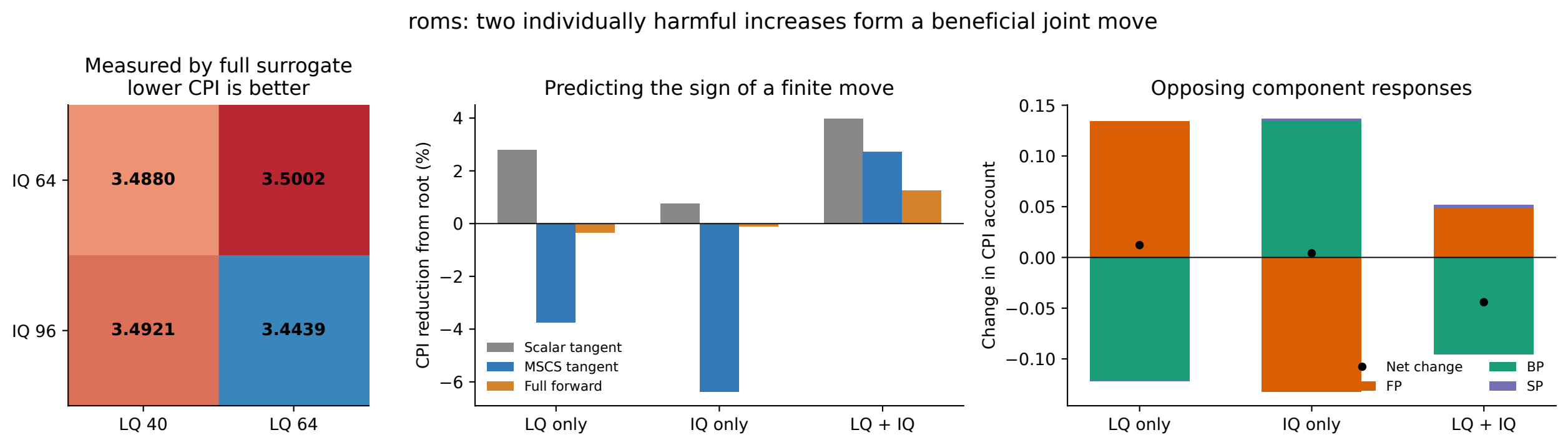
roms: two individually harmful increases form a beneficial joint move
Measured by full surrogate
lower CPI is better
IQ 64
IQ 96
3.4880
3.5002
3.4921
3.4439
LQ 40
LQ 64
Predicting the sign of a finite move
CPI reduction from root (%)
4
2
0
−2
−4
−6
Scalar tangent
MSCS tangent
Full forward
LQ only
IQ only
LQ + IQ
Opposing component responses
Change in CPI account
0.15
0.10
0.05
0.00
−0.05
−0.10
Net change
FP
BP
SP
LQ only
IQ only
LQ + IQ
Post-hoc diagnostic, frozen PRISM surrogate (not gem5). Stage accounts are averaged. No claim of universal search superiority.